CS 486/686
Sequence Models, Attention, and Transformers

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Lecture 19

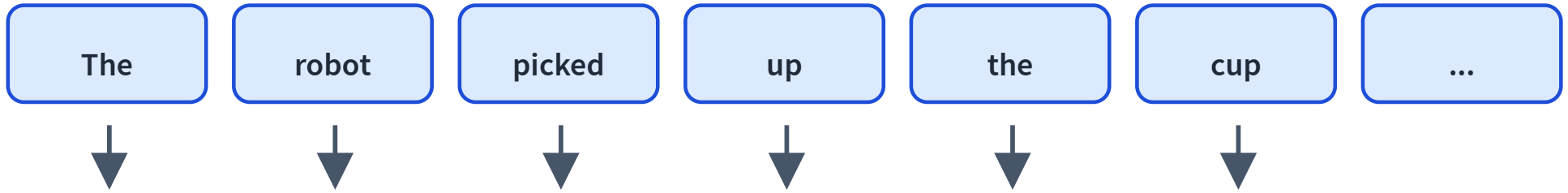
How embeddings talk to each other

Learning goals

- Model a sentence as a **sequence of embeddings**.
- Explain **RNNs** and why they struggle to scale.
- Explain **attention** as token-to-token retrieval.
- Assemble a **transformer block** and understand **causal masking**.

From embeddings to sequences

L18: each item can become a learned vector.



L19: how do these vectors interact?

A sentence is not a bag of words

dog bites man

man bites dog

Same words. Different order. Different meaning.

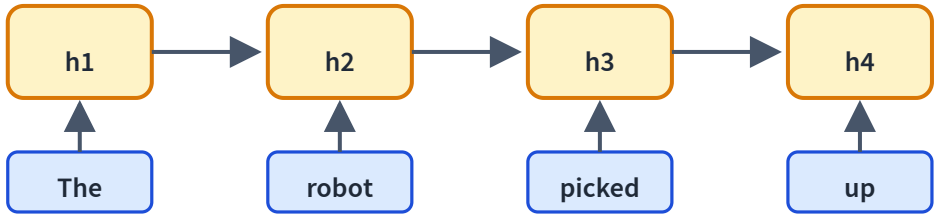
Running example

The robot picked up the cup because **it** was empty.

What does **it** refer to?

The answer depends on context, not the token alone.

The first idea: recurrent state



Read one token at a time.

$$h_t = f(h_{t-1}, x_t)$$

The hidden state summarizes the past.

RNN inference: update and pass forward

The	robot	picked	up
may encode: determiner	may encode: entity	may encode: action	may encode: phrase

The hidden state is one dense vector, not a literal list;
these labels are intuition for what may accumulate.

Why RNNs are limited

Sequential

Step t waits for $t - 1$.

Bottleneck

Everything squeezes into one state.

Long paths

Distant tokens interact through many steps.

Good concept, hard scaling story.

Modern recurrence revisits the idea

Structured state updates can be faster and stabler:

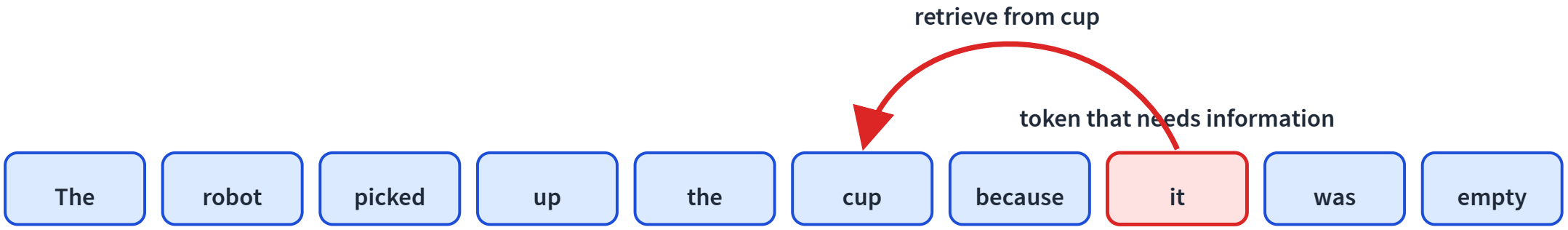
$$\text{state}_t = A\,\text{state}_{t-1} + Bx_t$$

> A is a learned state update; B maps the new input into state.

Modern state-space models keep recurrence, but redesign it for parallel hardware.

Different idea: let tokens look directly

When processing **it**, which earlier word contains useful information?



A decoder can look only at itself and earlier source tokens.

Reaching across the sequence

For an earlier source and a later query, the RNN path grows; attention stays one hop.

Interactive on the live slides: compare an earlier source with a later query. In an RNN, information crosses $n - 1$ sequential updates; causal attention lets the later query reach any earlier source in one hop, while computing $O(n^2)$ scores for the sequence.

Attention is content-based lookup

1 **Match**
Compare what *it* needs with each earlier source.

→

2 **Retrieve**
Bring back a weighted mix of source information.

query
the token doing the attending

key
each source token's address

value
that source token's payload

Keys decide where to look; values decide what comes back.

Queries, keys, values are learned projections

At the current layer, token i has hidden state $\mathbf{h}_i$:

need

$$\mathbf{q}_i = W_Q \mathbf{h}_i$$

query

address

$$\mathbf{k}_i = W_K \mathbf{h}_i$$

key

payload

$$\mathbf{v}_i = W_V \mathbf{h}_i$$

value

Same hidden state, three learned views. In deeper layers, $\mathbf{h}_i$ already contains context.

One query scores earlier keys

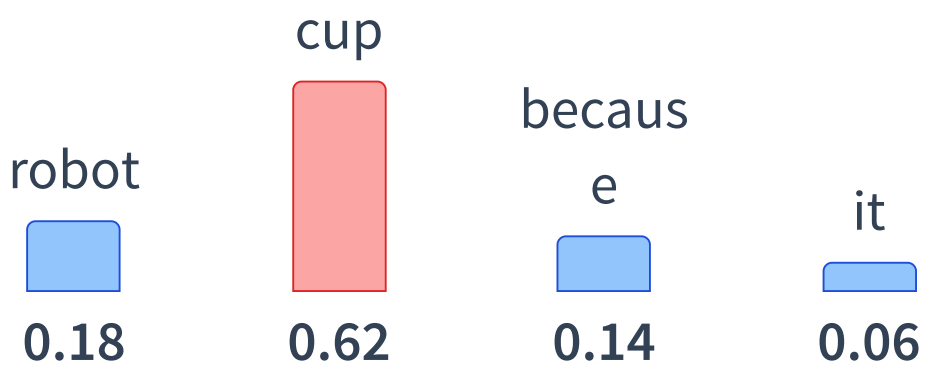
Current token: *it* supplies $\mathbf{q}_{\text{it}}$.

Allowed source j: $j \le i$; it supplies $\mathbf{k}_j$ now and $\mathbf{v}_j$ later.

$$\text{score}(\text{it}, j) = \mathbf{q}_{\text{it}}^{\top} \mathbf{k}_j$$

source key k_j	match score
robot	medium
cup	**high**
because	low
it (self)	low

Softmax turns scores into weights



The allowed source weights are nonnegative and sum to 1.

Use the weights to mix source values

$$\mathbf{z}_{\text{it}} = \sum_{j \le i} \alpha_{\text{it},j} \mathbf{v}_j$$

0.18 $\mathbf{v}_{\text{robot}}$	0.62 $\mathbf{v}_{\text{cup}}$
0.14 $\mathbf{v}_{\text{because}}$	0.06 $\mathbf{v}_{\text{it}}$

→

$\mathbf{z}_{\text{it}}$
new context vector

Because **cup** has the largest weight, its payload contributes most.

Scaled dot-product attention

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

$QK^\top$

all pairwise relevance scores

V

the information being mixed

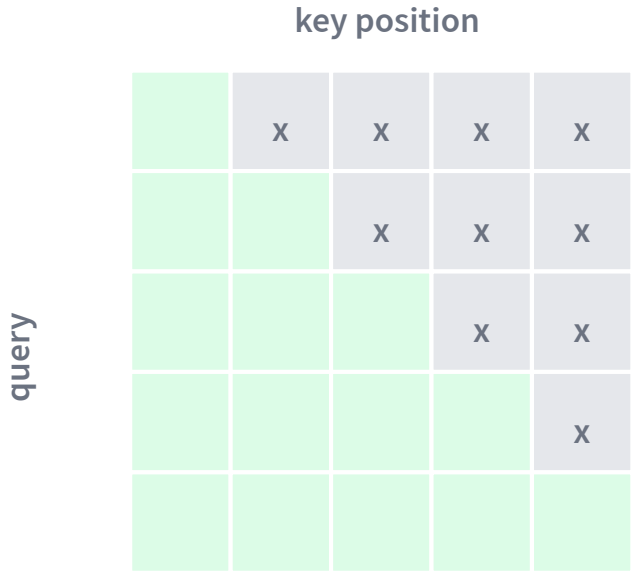
d_k is the query/key width; dividing by $\sqrt{d_k}$ keeps large dot products from saturating softmax.

Scores → weights

Softmax turns raw scores into attention weights — scaling sharpens or flattens them.

Interactive on the live slides: drag the scores and the temperature and watch softmax turn them into a probability distribution. Low temperature (like large scores, or a small $\sqrt{d_k}$) sharpens attention onto one token; high temperature flattens it toward a uniform average.

Causal masking prevents peeking



A decoder predicts the next token from the prefix.

Query position i may use keys/values only from $j \leq i$.

$$\text{softmax}\left(\frac{QK^\top + M}{\sqrt{d_k}}\right)V$$

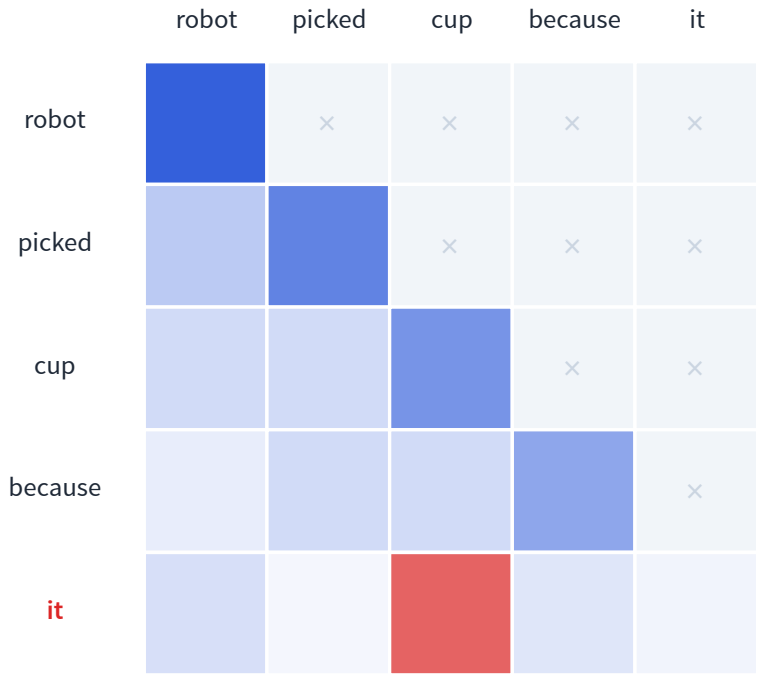
$M_{ij} = 0$ when $j \leq i$
$M_{ij} = -\infty$ when $j > i$

Causal masking, interactively

Toggle the mask and pick a query word — watch the weights renormalize over allowed positions.

Interactive on the live slides: toggle the causal mask on and off and click a query word. With the mask on, every future cell is blocked and that token's attention renormalizes over positions $\leq t$ only; with it off, the token attends both directions. This one switch is what turns a transformer into a left-to-right language generator.

Every row asks where one query looks



Rows: query positions.

Columns: source keys/values.

Toy row: query **it** puts its largest weight on **cup**.

Gray cells are future positions blocked by the causal mask.

Real attention in a decoder

Actual Qwen3-0.6B causal attention — pick a head, then click a query word.

Interactive on the live slides: real attention from Qwen3-0.6B, the small decoder model dissected in L21. Future cells are causally masked. Pick a measured layer/head pattern and click a query word to inspect its attention row; one selected head gives high weight from **it** to **cup**, another emphasizes the previous token, and another spreads over the allowed prefix. Attention is inspectable, but is not automatically a faithful explanation.

Multi-head attention repeats the lookup

Each head learns different W_Q, W_K, W_V projections.

source link

high *it* → *cup*

local pattern

previous token

broad pattern

spread over prefix

Heads can emphasize different patterns in parallel; the real demo shows measured examples.

Transformers dropped recurrence—where did order go?

RNN

Order is built into sequential updates.

Transformer, layer 1

all token vectors processed in parallel

Content alone does not label first, second, third…

Fix the final query and shuffle the same previous vectors:
content-only first-layer attention is unchanged *by word*.

Position changes the attention scores

simple start: $\mathbf{h}_i^0 = \text{token}_i + \text{position}_i$

Now the projected queries and keys depend on both content and slot.

Modern decoders: Qwen/GPT-style models commonly rotate query/key dimensions with RoPE, encoding relative position directly in their match.

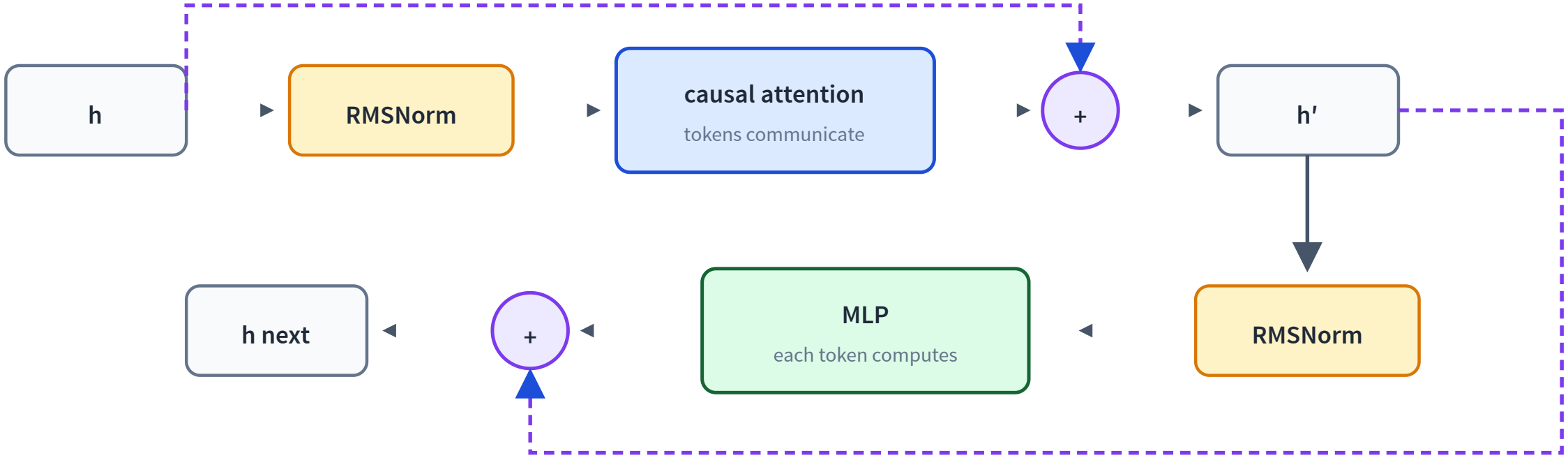
Mask: earlier vs future. **Position signal:** richer relative/absolute order.

Shuffle the past: what changes?

Keep the final query fixed; shuffle only the same previous words.

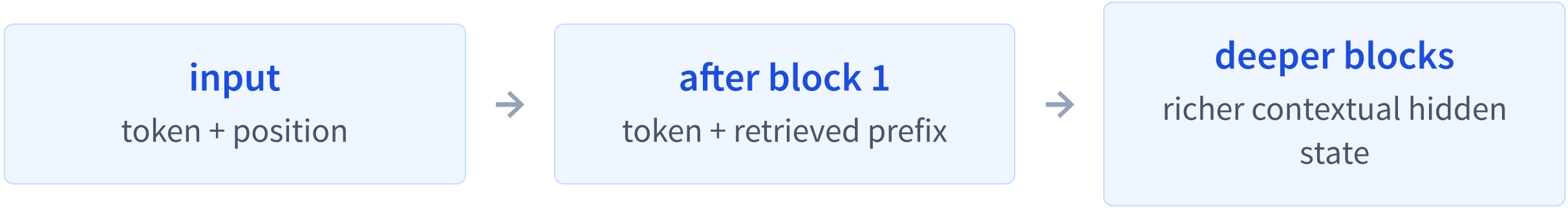
Interactive on the live slides: compare two prefixes containing the same words in different orders, followed by the same final query. In the first layer, content-only attention assigns each word the same weight after the shuffle; adding position information changes the scores. Important deeper-layer caveat: previous hidden states may already differ because each word was contextualized from a different causal prefix, so later-layer keys and values can change even without an explicit position embedding.

A decoder block: communicate, then compute



Residual paths preserve the old representation; normalization stabilizes scale. Exact variants differ—Qwen uses a pre-norm/RMSNorm-style block.

Keys and values become contextual



Shuffle previous words and their causal histories change—so deeper-layer $\mathbf{k}_j, \mathbf{v}_j$ can change too.

This is why the first-layer “same set” result is useful but not the whole transformer story.

Two common attention masks

Bidirectional encoder

Every token may use both left and right context. BERT-style.

Causal decoder

Position i uses only $j \leq i$. Qwen/GPT-style.

Today's main path is the decoder: its mask enables next-token training and generation.

Why transformers took over

Parallel

All positions at once during training.

Long-range

Any allowed source is one attention hop away.

Scalable

Bigger models and data kept improving.

Now we have the architecture

Next we need the training task:

predict the next token at massive scale

L20: pretraining language models from first principles.

Recap

✓ RNNs summarize the past with a **hidden state**.

✓ Attention retrieves information directly between **tokens**.

✓ Queries, keys, and values implement learned retrieval.

✓ Transformer blocks stack **attention** and **feed-forward** computation.

✓ Causal masks enable autoregressive training and generation.